

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. They need to understand what the actual problem is. They will still have to solve and work out what the problem needs.
2. They will need to design, test and verify to make sure everything works and the problem is solved
3. Ethics plays a big part into this and privacy aswell.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A 50-line calculator	Yes	Not really	It's coding with little to none engineering process.
B Payroll for 5000 employees	Yes	Yes	There are many users, and actual requirements, testing and errors could have real consequences
C AI generates an appointment app from one prompt	Yes	No	

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Patients	Easy booking and cancelling, accurate appointment details
GPs	Clear view of schedule and availability and quick access to patient information
Admin	Fast patient search, consistent appointment status
Clinic management	Reliable operational reports, few errors, simple system

Task 2 - Identify current problems

1. Duplicate appointment booking
2. Patient records are hard to find
3. Appointment status is inconsistent
4. Practitioner availability isn't visible

Task 3 - Ask client questions

1. What information must be stored for patients and appointments
2. Which reports does management need
3. How many practitioners, staff and patients and daily appointments do clinics have
4. What statuses and booking rules exist
5. Is booking done only by staff or also patients

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes	Yes	Include
Facial recognition login	No	No	Reject
AI diagnosis recommendations	No	No	Reject
Patient search	Yes	Yes	Include
Online payment	No	No	Reject
Practitioner schedule view	Yes	Yes	Include
Insurance processing	None	No	Reject

Treatment-plan generation	No	No	Reject
---------------------------	----	----	--------

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

Testing. Validation requirements with other stakeholders. This confirms what clinic really needs and doesn't need AI at all.